AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter Monitoring & Health Effects

Issue 2026-07-28 | Entry window 28 July 2026 (PubMed EDAT); Europe PMC creation 26–28 July | 21 screened, 19 in scope

19 RECORDS IN SCOPE (2 EXCLUDED)	**10** SUBTOPIC CLUSTERS ALL REPRESENTED	**8** RECORDS FROM CHINA (42% OF CORPUS)	**2** ESTIMATES REPORTED WITH A 95% CI	**12** RECORDS IN MDPI TITLES (63%)

Signal of the day

The binding constraint on African exposure epidemiology is record linkage, not instrumentation

The AIRCLARE-Ghana pilot (Research Square preprint, `rs-10441248`) is the first time-stratified case-crossover study of short-term particulate exposure in Ghana. Two years of *federal-equivalent-method* $PM_{2.5}/PM_{10}$ monitoring were linked to 213,241 outpatient records at the University of Ghana Hospital, yielding 9,511 acute respiratory presentations. Each IQR rise in $PM_{2.5}$ (14.70 $\mu g/m^3$) raised the odds of an upper-respiratory presentation by **3.8% (OR 1.038, 95% CI 1.010–1.068)**; PM_{10} (IQR 51.49 $\mu g/m^3$) gave **OR 1.030 (1.008–1.052)**. The lag structure peaks at day 1 (OR $\approx$ 1.07 / 1.06) and persists to a cumulative 0–7 d (OR $\approx$ 1.09 / 1.06), with the strongest signal during the dry Harmattan season and among women.

Two things matter here for a distributed-sensing programme. First, the reported **96.03% monitoring-stream completeness after QC** is the study's real contribution: it shows the reference-grade data layer is not the failure point in a low-resource deployment — the pipeline that joins it to clinical records is. Second, the effect sizes are small enough that they are only recoverable because exposure misclassification was minimised by a single, co-located, high-completeness monitor serving a geographically compact catchment. Any attempt to scale this design across Accra with a distributed low-cost network will face the opposite trade-off: better spatial resolution, worse and drift-dependent measurement error. Pneumonia and asthma associations were directionally consistent but null, which is what an underpowered pilot at these effect sizes should produce, and the authors say so.

Caveats stated plainly: this is a **preprint**, not peer reviewed; the catchment is a university hospital and student clinic, so the population skews young, educated and urban and the 925.7-per-1,000 URI rate is not a community incidence; a single monitoring site is treated as the exposure for all subjects, so within-city gradients are absorbed into error; and the six-stage ICD-10 filtering cascade is a bespoke, unvalidated case definition whose sensitivity/specificity are unreported. The case-crossover design does control time-invariant confounding by construction.

What changed today

- **Sub-Saharan Africa enters the corpus.** Yesterday's 33-record issue contained zero records from the region; today two of nineteen (Ghana case-crossover; Uganda scoping review) originate there. Both are explicitly about monitoring infrastructure rather than biology, which is the correct order of operations for a region with essentially no long-term reference network.
- **Quantitative reporting quality collapsed.** Yesterday twelve effect estimates with intervals were extractable from 33 records. Today **two of nineteen** records report a point estimate with a 95% CI. Several report bare point estimates (PD-L1 OR $= 1.035$, $p = 0.017$), ranges across unspecified model sets (sleep disorders, "OR 1.08 to 3.20"), or standardised regression coefficients with significance stars only (spatial Durbin, 0.028*). This is not a sampling artefact — it tracks the journal-composition shift below.
- **Journal composition shifted sharply.** Twelve of nineteen records (63%) are in MDPI titles (*Toxics* alone contributes seven). Reference-grade cohort work in *JAMA Netw Open*, *J Am Heart Assoc* and *Diabetologia*, which carried yesterday's issue, is absent. Weight the corpus accordingly: today's tier-A count is three, against nine yesterday.
- **Instrumentation produced nothing.** Zero records in this entry window concern low-cost sensor characterisation, co-location calibration, drift correction, network siting, or satellite/model fusion. The *Sensing* cluster is filled by a

bioaerosol surveillance campaign and a bibliometric review. A supplementary sweep found relevant 2026 work (e.g. AMT 19, 1293) but it falls outside the entry window and belongs to no issue of this digest. The gap is real, not a query failure: PubMed does not index *Atmos. Meas. Tech.* or *Atmos. Chem. Phys.*, Europe PMC returned no creations for 28 July, and OpenAlex has moved its `created_date` filter behind a paid plan — see *Provenance*.

- **Microenvironmental measurement is the one growth area.** Five of nineteen are measurement campaigns and all five are indoor or near-source: subway platforms under different screen-door geometries, charcoal-barbecue dining rooms, an urban supersite, and SARS-CoV-2 RNA recovered from ambient $PM_{2.5}$ filters. The common structure is a fixed sampler in a small, high-concentration volume — exactly the regime where a calibrated low-cost node would be adequate and a reference sampler is overkill.

Corpus analytics

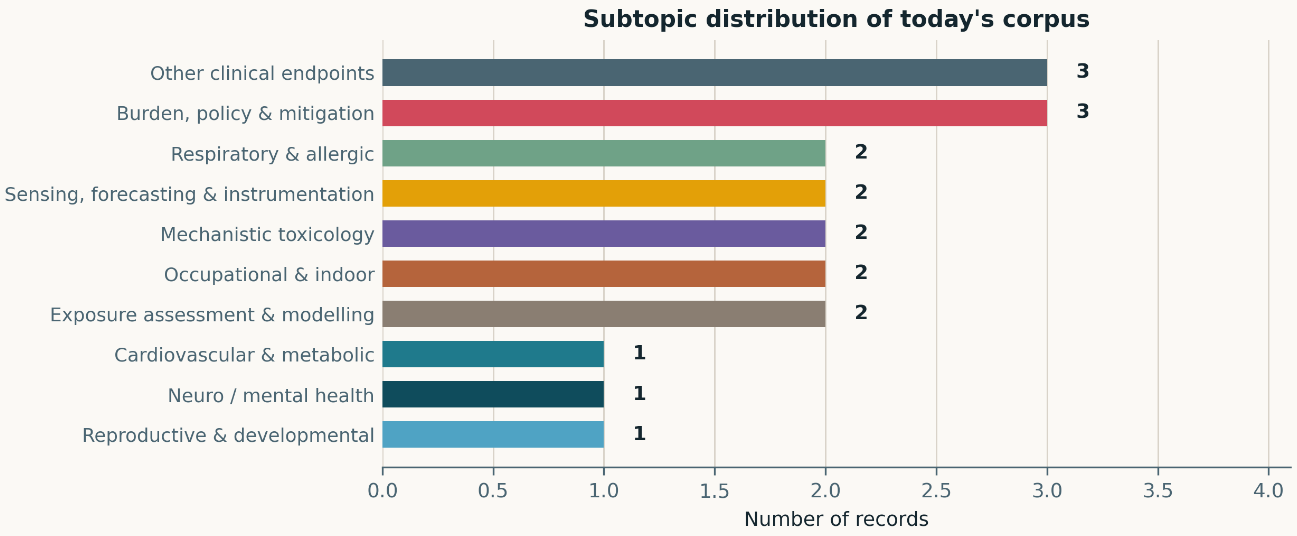
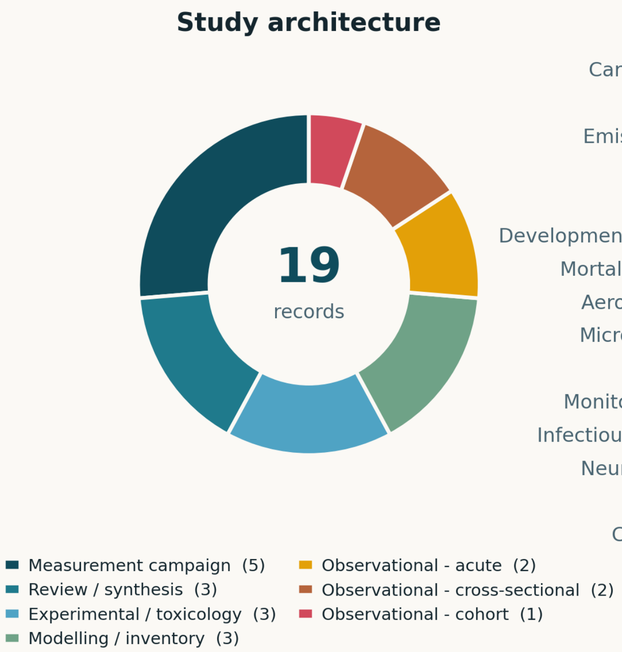


Fig. Subtopic assignment of the 19 in-scope records; single-label by dominant contribution. Unlike yesterday's issue there is no dominant cluster: *Burden, policy & mitigation* and *Other clinical endpoints* lead with three each and six clusters carry one or two. A flat distribution across a small n carries little information about the field — it mostly reflects that one publisher's mid-month batch supplied most of the corpus.



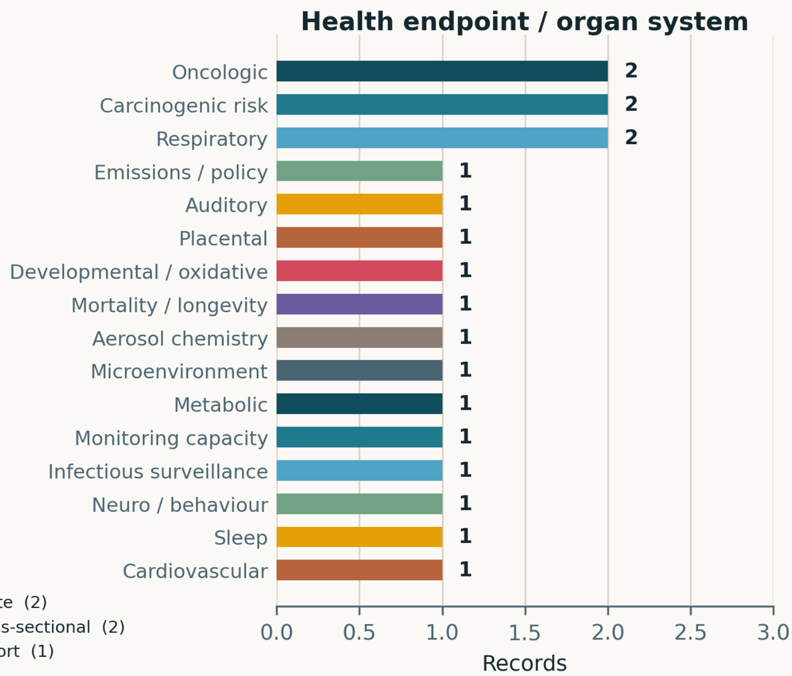


Fig. Left — study architecture. Measurement campaigns are the largest single class (5/19, 26%), a reversal of the review-heavy 27 July issue. Only one prospective cohort appears, and no randomised or controlled-exposure human study. Right — endpoint by organ system. Oncologic, carcinogenic-risk and respiratory endpoints tie at two; the long tail of single-record endpoints (sleep, auditory, placental, microenvironment, aerosol chemistry) is characteristic of a corpus assembled from a general-toxicology journal issue rather than from a coherent research front.

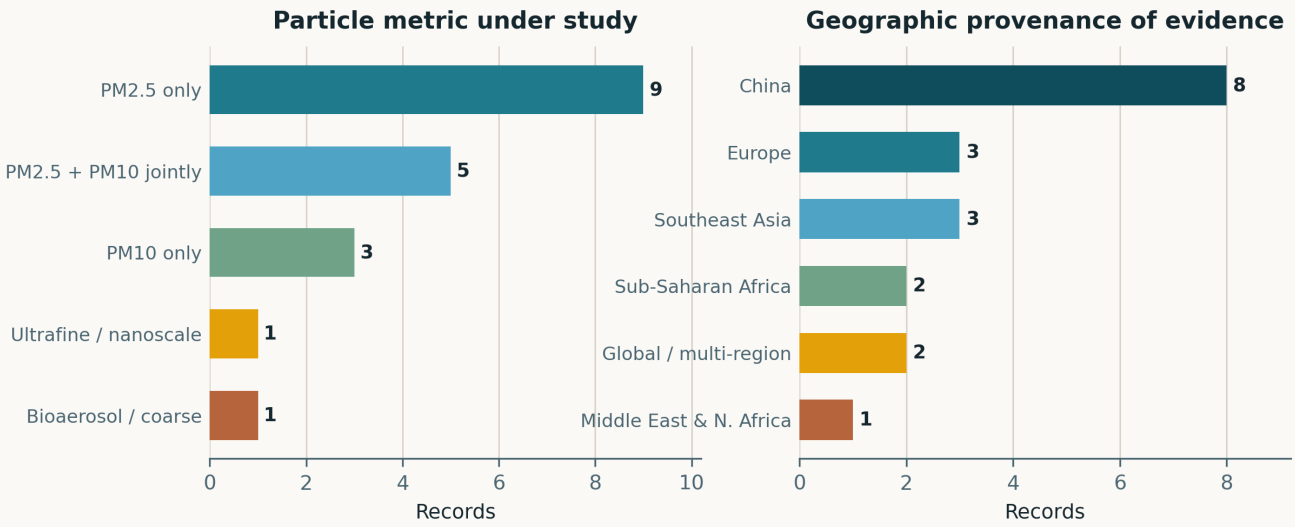


Fig. Left — particle metric. $PM_{2.5}$ alone is the exposure in nine records and appears jointly with PM_{10} in five more. Ultrafine particles are again the explicit subject of exactly one record, and that record measures no particles at all (it is a spatial-econometric inference about UFP sources from industrial-output statistics). Right — geographic provenance. China supplies eight (42%), Southeast Asia three (all Thailand, all Chiang Mai haze-season work), and sub-Saharan Africa two. No record from South Asia.

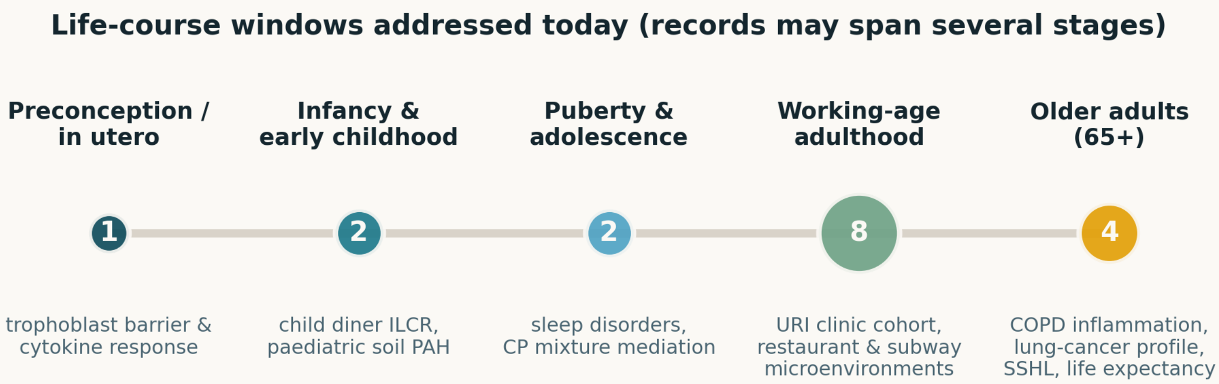


Fig. Life-course windows addressed; counts are non-exclusive and reflect the population actually studied, not the population the authors generalise to. The mass sits in working-age adulthood (8) because today's measurement campaigns sample occupational and commuter microenvironments. The in-utero window, dominant yesterday at five records, drops to one — and that one is a cell-line experiment in which the PM arm was null.

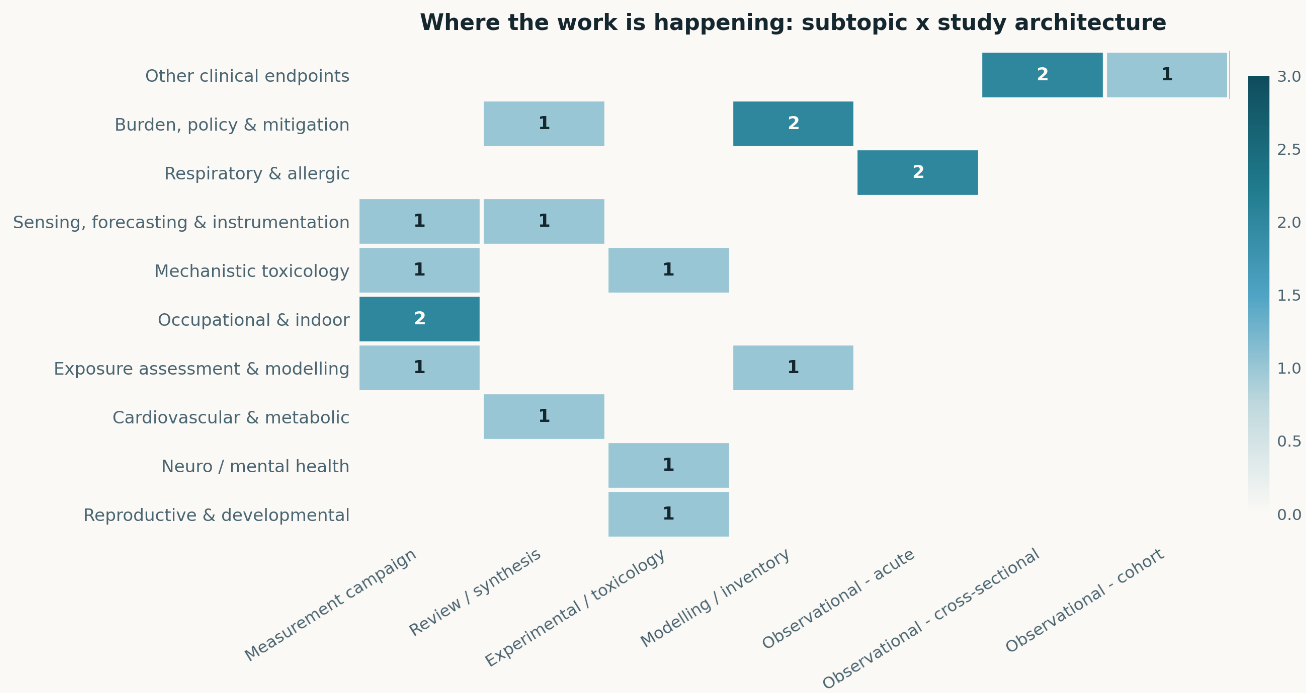


Fig. Subtopic × study-architecture cross-tabulation. The diagnostic feature is how empty it is: 19 records occupy 14 of 70 cells, with no cell above two. Note the *Sensing* row — one measurement campaign, one review, and no modelling and no linkage to any health endpoint. The *Occupational & indoor* row is purely measurement, with no exposure–response analysis attached to either campaign. The instrumentation/epidemiology silo flagged on 27 July persists unchanged.

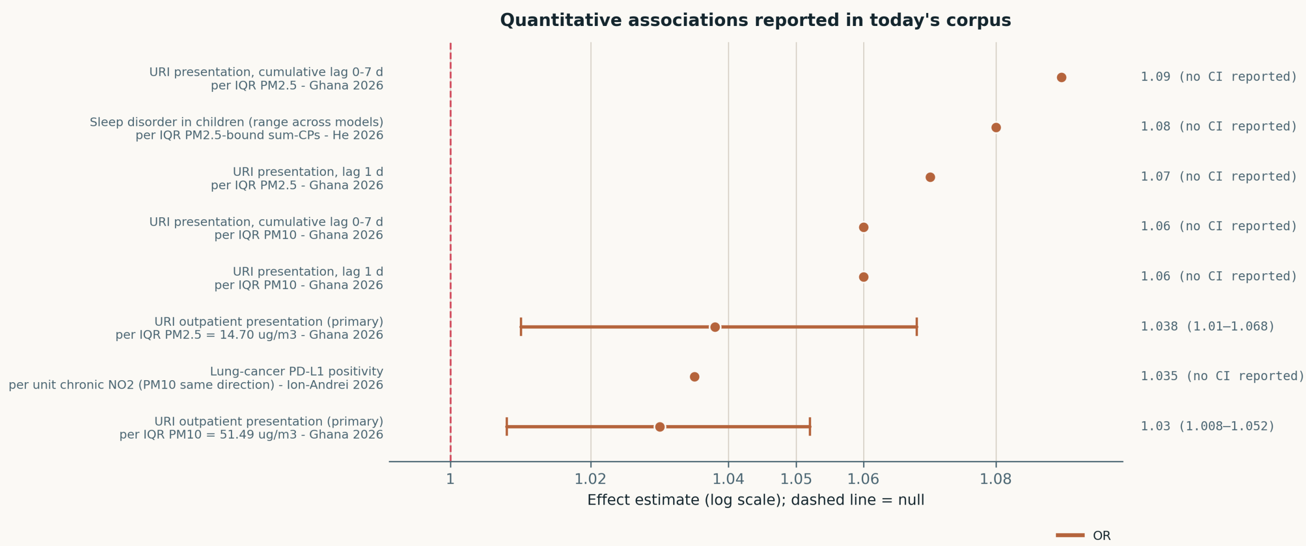


Fig. Every quantitative association extractable from today's corpus, on a common logarithmic axis. **Only the two bottom-most estimates carry a reported interval**; markers drawn without whiskers are point estimates for which the source published no CI, and they should be read as unfalsifiable until the full text is available. Note also the axis range: all estimates lie between 1.03 and 1.09, an order of magnitude tighter than yesterday's spread (1.03–3.16), because today's corpus contains no high-contrast exposure comparison and no rare-outcome registry study. The "sleep disorder" marker is the *lower bound* of a reported OR range (1.08–3.20) whose model set is unspecified in the abstract; it is plotted at that bound deliberately.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core, the finding that matters, the limitation that actually threatens the inference, and the read-across to distributed low-cost PM sensing.

Neuro, cognition & mental health 1 record

Lu et al. 2026b Environ Res

Subchronic $PM_{2.5}$ exposure exacerbates chronic stress-induced depression-like pathology in mice

Factorial murine design crossing subchronic $PM_{2.5}$ inhalation with chronic unpredictable mild stress (CUMS), read out by behavioural batteries, hippocampal histopathology, transcriptomics, electrophysiology and synaptic-plasticity markers. The result that matters is the **interaction, not the main effect**: $PM_{2.5}$ alone produced no depression- or anxiety-like behaviour, only modest neurobiological change, whereas $PM_{2.5}$ + CUMS worsened anhedonia, behavioural despair and anxiety relative to CUMS alone, with greater hippocampal neuroinflammation, redox disruption and impaired synaptic plasticity. This supports an effect-modifier rather than a direct-cause model. The abstract reports no exposure concentration, no particle characterisation and no effect sizes, so dose–response is unassessable and the inhalation regime cannot be compared with ambient exposure; "real-ambient" status is not claimed. For sensing, the implication is that a monitoring design aimed at mental-health endpoints needs to resolve *co-occurrence* of exposure and psychosocial stressor, which is a personal-monitoring problem, not a fixed-site one.

doi: 10.1016/j.envres.2026.125327

Cardiovascular & metabolic 1 record

Zhou et al. 2026 iScience

Air pollution and hypertension: an umbrella review of systematic reviews and meta-analyses

Umbrella review over three databases to September 2025, AMSTAR-2-appraised, retaining eleven meta-analyses (eight on essential hypertension, three on gestational hypertension). Conclusion: cumulative evidence supports $PM_{2.5}$ and PM_{10} raising essential hypertension risk, while the gestational-hypertension association remains inconclusive on three studies. The value of this record is as a citation anchor and a map of between-review heterogeneity; the abstract reports **no pooled estimate, no I^2, and no AMSTAR-2 distribution**, so the strength of the "plausible role" claim cannot be checked without the full text. Umbrella reviews also inherit the primary literature's exposure-model heterogeneity — land-use regression, satellite-derived and nearest-monitor assignments are pooled as though interchangeable, which is precisely the assumption a co-located sensor network is positioned to test.

doi: 10.1016/j.isci.2026.116811

Reproductive & developmental 1 record

Liu et al. 2026 Int J Mol Sci

Deoxynivalenol, cytokines and particulate matter in human trophoblast cell models

Head-to-head sensitivity comparison of HTR-8/SVneo and BeWo trophoblast lines against three maternal-exposome stimuli — the mycotoxin deoxynivalenol, a TNF-α/IL-1β/ IFN-γ cocktail, and $PM_{2.5}$ — at two penicillin/streptomycin concentrations, 24 h exposure. **The PM arm was null**: no change in IL-6, IL-8 or hCG in either line, while DON and the cytokine cocktail moved IL-6/IL-8, hCG and junctional markers (ZO-1, OCLD, CLDN-3/4, E-CAD). A negative result in a 24 h submerged monoculture at an unreported dose is weak evidence of placental safety and strong evidence that these two lines are the wrong model for particle toxicity — there is no air–liquid interface, no deposition physics and no particle characterisation. Worth logging precisely because prenatal-PM epidemiology keeps citing trophoblast inflammation as its mechanism.

doi: 10.3390/ijms27146434

Respiratory & allergic 2 records

AIRCLARE-Ghana 2026 (preprint) Research Square

Short-term ambient $PM_{2.5}/PM_{10}$ and acute respiratory morbidity in Ghana

See *Signal of the day*. Time-stratified case-crossover, Jan 2023–Dec 2024, 9,511 acute respiratory presentations from 213,241 outpatient records, conditional logistic regression adjusted for temperature and relative humidity. **URI: OR 1.038 (1.010–1.068) per IQR $PM_{2.5}$; OR 1.030 (1.008–1.052) per IQR PM_{10}**, peaking at lag 1 and persisting to cumulative lag 0–7 d. Pneumonia null, asthma null. Threat to inference: one monitoring site stands in for a whole catchment, so Berkson-type and classical exposure error coexist and the true effect is plausibly larger; the case definition is bespoke and unvalidated; and it is a preprint. The 96.03% post-QC completeness figure is the number a network operator should take away.

doi: 10.21203/rs.3.rs-10441248/v1

Chaiwong et al. 2026 Int J Mol Sci

$PM_{2.5}$ and systemic inflammatory biomarkers in elderly COPD versus healthy elderly

Pilot panel of 38 COPD patients (73.6 ± 7.1 y) and 20 age-matched controls, sampled in a high- (79.1 ± 23.3 $\mu g/m^3$) and a low-pollution (13.9 ± 3.6 $\mu g/m^3$) Chiang Mai season, measuring hsCRP, IL-6, IL-8 and TNF-α. IL-8 was higher in COPD across both periods, but **no significant**

within-group increase accompanied the 5.7-fold seasonal exposure contrast — a null the authors attribute to inhaled corticosteroids (65.8% of COPD) and statins (42% COPD, 65% controls). That explanation is untestable at $n = 58$; equally consistent are ceiling effects on chronic inflammation, ambient-monitor exposure misclassification, and simple lack of power. Reported as a pilot, and should be cited as one. Relevance: a seasonal contrast this large failing to move systemic biomarkers argues for personal rather than municipal exposure metrics in susceptible-population panels.

doi: 10.3390/ijms27146283

Mechanistic toxicology 2 records

Fakfum et al. 2026 Toxics

Seasonal $PM_{2.5}$ exposure and plasma metabolome changes related to metabolic syndrome

Prospective within-subject comparison of plasma ^{1}H-NMR metabolomes in 25 healthy adults in Samoeng District, Chiang Mai, between haze and non-haze seasons. Twenty-six metabolites separated the seasons by PLS-DA (VIP > 1.5): eleven up (maleylacetoacetic acid, deoxyribose 5-phosphate, betaine, 3-hydroxyanthranilic acid, 1-methyladenosine) and fifteen down (deoxyguanosine, D-arabitol, glycerophosphocholine, ophthalmic acid, oxaloacetic acid), implicating inflammation, ROS generation, NO inhibition and disrupted energy metabolism across 56 pathways. The design cannot separate $PM_{2.5}$ from everything else that changes with the Thai dry season — temperature, humidity, diet, activity, and co-emitted gases — and PLS-DA at $n = 25$ over hundreds of features is prone to overfitting without the permutation testing the abstract does not report. Hypothesis-generating.

doi: 10.3390/toxics14070544

Lu et al. 2026 Toxics

Toxicity of $PM_{2.5}$ from incomplete solid-fuel burning in Caenorhabditis elegans

Synchronised L4 *C. elegans* exposed to aqueous suspensions of $PM_{2.5}$ collected from incomplete combustion of six solid fuels (rice, wheat, peanut and rapeseed straw; poplar and paulownia branch), scored for body length, fertilised-egg count, lipofuscin accumulation and ROS, alongside OC/EC, water-soluble ion and PAH characterisation. **Oilseed-crop straw (peanut, rapeseed) was the most toxic**, and the effect tracked lipid-soluble PAH content via oxidative stress. The source-specificity result is the useful part — it says fuel type, not mass concentration, drives toxicity, which is an argument for speciation-capable monitoring. The model is the limitation: an aqueous suspension delivered to a nematode cuticle and gut has no relationship to pulmonary deposition, and no mammalian or human-cell confirmation is offered.

doi: 10.3390/toxics14070597

Occupational & indoor exposure 2 records

Komonnithiphong et al. 2026 Toxics

$PM_{2.5}$-bound PAHs and carcinogenic risk in charcoal barbecue (Moo Kratha) restaurants

Ten area samples taken at customer dining tables in ten Chiang Mai charcoal-barbecue restaurants during peak evening hours in February 2026 (early dry season), 2 L min^{-1} for 180 min onto quartz fibre filters, 16 US-EPA priority PAHs by GC-MS. Mean $PM_{2.5}$ **87.64 μg m^{-3}**, total PAHs 132.74 ng m^{-3}, carcinogenic fraction 52.3%, TEQ_{BaP} 15.34 ng m^{-3}. Inhalation ILCR for a frequent adult diner **2.82×10^{-5}** (child 2.12×10^{-5}), rising to 6.16 and 4.62×10^{-5} at the worst site; even monthly diners exceed 10^{-6}. Two hard limitations: $n = 10$ single visits with no repeat sampling gives no within-site variance, and February in Chiang Mai means the regional haze background is confounded with the cooking source — no outdoor or upwind reference was reported. A paired indoor/outdoor low-cost node would have resolved that for a fraction of the cost.

doi: 10.3390/toxics14070643

Subway PSD study 2026 J Environ Sci (China)

PM and CO_2 in Chinese subway stations with varying platform screen door forms

Seven stations instrumented across platform, hall and outdoor locations, contrasting full-height platform screen doors (PSD) with half-height platform bailout doors (HPBD). Platform > hall > outdoor for PM; CO_2 tracks passenger count linearly. The piston effect on train arrival raises **$PM_{2.5}$ by 20% and PM_{10} by 36% under HPBD**, appreciably more than under PSD, while CO_2 falls 1.8% (PSD) and 4.6% (HPBD). Fine particles carry 38.22–62.57% of total mass, highest behind PSD, and Fe dominates the elemental profile across six apportioned sources. The design is a snapshot without seasonal or diurnal repetition, and the door form is confounded with station age, depth and ventilation design, so the PSD-versus-HPBD contrast is associational. Still the cleanest demonstration in today's corpus that a metric-and-geometry-specific enclosed microenvironment needs its own sensor siting logic — a platform-edge node and a concourse node are measuring different populations of particles.

doi: 10.1016/j.jes.2025.12.027

Sensing, forecasting & exposure modelling 2 records

Wang et al. 2026 Viruses

Joint monitoring and early warning of SARS-CoV-2 in outdoor $PM_{2.5}$ and wastewater

Twelve months (2023) of paired outdoor $PM_{2.5}$ filter and wastewater sampling in Fuzhou, RT-qPCR for SARS-CoV-2 RNA, associations with pollutants and meteorology by GAM and DLNM. **45.9% of $PM_{2.5}$ samples were positive**; viral load in $PM_{2.5}$ rose with CO, $PM_{2.5}$ mass and atmospheric pressure and fell with temperature and sunshine duration, with the **strongest lead over outpatient COVID cases at ≈ 2 days** against lag 0 for wastewater. The direct interest for a distributed-sensing group is that a routine gravimetric filter stream doubles as a bioaerosol surveillance substrate with a two-day lead — but only if filters are archived and chain-of-custody preserved, which most low-cost optical nodes cannot do at all. The weakness is that RNA detection on a filter conflates aerosol transmission with resuspended or deposited material, and no infectivity assay was attempted; the "early warning" claim rests on ecological time-series correlation with outpatient counts.

doi: 10.3390/v18070788

Ainembabazi et al. 2026 Toxics

Trends in atmospheric pollution research in Uganda, 1990–2025: a scoping review

Structured scoping review across Google Scholar, Web of Science and PubMed. Output was negligible before 2010, expanded after 2015 and surged 2020–2025. $PM_{2.5}/PM_{10}$ dominate throughout while NO_2, SO_2, CO and O_3 are underrepresented; studies cluster in urban Kampala with almost no rural ambient monitoring; and the methodological trajectory runs from proxy and gravimetric approaches to **low-cost sensors, portable monitors and satellite-derived products**. The authors' own conclusion is the operative one: short-term, spatially constrained campaigns predominate, leaving no long-term or nationally representative record. This is the clearest statement in today's corpus of the niche a sustained calibrated low-cost network fills — and a reminder that Google Scholar inclusion makes the sampling frame non-reproducible, so the counts should be treated as indicative.

doi: 10.3390/toxics14070542

Pająk et al. 2026 Toxics

Toxicological legacy of PAHs from an urban tire fire: soil contamination and cancer risk

Source-to-receptor assessment of a large landfill tire fire coupling HYSPLIT dispersion modelling against air-quality monitor data with soil sampling on the fireground and in surrounding allotment gardens, screened for Zn as a tire tracer and 16 priority PAHs. Emissions **PM_{10} 1.34 t and PAHs 17.7 kg**; fireground soil ΣPAH 148.9 mg kg^{-1}; cumulative child ILCR **9.7×10^{-4}**, an order of magnitude above the 10^{-4} regulatory benchmark, with dermal contact rather than inhalation the dominant pathway and elevated risk (10^{-5}–10^{-4}) persisting at distal residential sites. Ecological risk quotients exceeded unity for PAHs everywhere and for Zn/Cu near the fire. Limitation: no pre-fire soil baseline, so urban background PAH — itself substantial in a coal-heating region — is attributed to the fire; and HYSPLIT at this scale is validated against a sparse monitor network. The transferable point is that acute source events deposit a multi-year soil legacy that airborne monitoring alone will not detect.

doi: 10.3390/toxics14070543

Chengdu supersite 2026 J Environ Sci (China)

Gas-phase reactions accelerate nitrate formation in spring and early summer, Sichuan Basin

Urban supersite speciation of $PM_{2.5}$ over April–June 2018–2021 in Chengdu, targeting the under-studied warm-season nitrate problem. During complex $PM_{2.5}/O_3$ episodes nitrate rose **47.8%, from 5.52 to 8.16 μg m^{-3}**, with its share of $PM_{2.5}$ climbing from 13.4% to 22.2%. The $NO_3^-/PM_{2.5}$ ratio scales with aerosol liquid water, and rapid early-morning ALW growth drives hygroscopic growth and heterogeneous production on top of the gas-phase channel. Directly consequential for optical sensing: a nitrate- and water-dominated aerosol has a very different hygroscopic growth factor and refractive index from the winter carbonaceous mixture most low-cost sensor calibrations are fitted to, so a humidity correction trained in winter will bias warm-season estimates. Observational and single-city, so the mechanism is inferred from covariation rather than tested.

doi: 10.1016/j.jes.2026.02.002

Burden, policy & mitigation — 3 records

Hu et al. 2026 Front Public Health

Older-adult life expectancy in China: Bayesian mortality estimation and spatially heterogeneous determinants

Two-stage design on the 2015 China 1% Population Sample Survey: Bayesian random-effects smoothing of age-specific mortality for 60+ across 31 provinces, abridged life tables, then a bake-off of MLR, spatial error, spatial lag, GWR and MGWR for socio-environmental determinants. **PM_{10} was not significant in any global model**; only GWR suggested a more negative association in south-western China, and GWR outperformed MGWR (R^2 0.789 vs 0.664), which the authors correctly attribute to bandwidth instability at $n = 31$. Read this as a negative result about method, not about PM: 31 aggregated provincial units, one cross-section, provincial mean PM_{10} as the exposure, and no lag structure cannot recover a pollution–longevity gradient that individual-level cohorts detect routinely. The paper is a clean illustration of how ecological aggregation destroys an exposure contrast that a denser monitoring grid would preserve.

doi: 10.3389/fpubh.2026.1851714

Zhou et al. 2026b Toxics

Sources of ultrafine particles during severe haze periods: a spatial Durbin analysis

Spatial Durbin model over the 28 "2+26" Beijing-Tianjin-Hebei channel cities, relating pollution-intensive industrial activity to haze, and attributing ultrafine particle generation to desulfurisation, denitrification and dust-removal operations — specifically lax wet-FGD management, ratcheting NO_x standards, and electrostatic precipitators that are ineffective in the ultrafine size range. Direct effect 0.028, total spillover-inclusive effect 0.151. **No ultrafine particles were measured anywhere in this study.** The UFP attribution is inferred from city-level industrial statistics and a haze proxy, so the central claim — that abatement equipment is itself a UFP source — is an untested hypothesis dressed as a finding, however physically plausible the ESP argument is. Logged tier C for the hypothesis, not the evidence; it is exactly the claim a size-resolved mobility-sizer campaign near a controlled stack could settle.

doi: 10.3390/toxics14070588

Sun et al. 2026 Br J Cancer

Clarifying causal and translational inference in studies of ambient $PM_{2.5}$ and lung cancer

Correspondence, not primary research, arguing that the $PM_{2.5}$–lung-cancer literature conflates aetiological causal claims with translational or prognostic ones, and that the distinction changes what an estimate licenses. No new data. Worth keeping in the register because the same conflation runs through exposure-assignment practice: a satellite- or LUR-derived annual mean supports population-attributable-fraction arithmetic but not the individual-level counterfactual that translational framing implies, and adding a denser sensor network narrows the measurement error without touching the identification problem.

doi: 10.1038/s41416-026-03555-2

Other clinical endpoints 3 records

He et al. 2026 Toxics

$PM_{2.5}$-bound chlorinated paraffins and sleep disorders in youth, with body weight as mediator

Largest study in today's corpus: 122,965 valid questionnaires from school-aged children across the Pearl River Delta, with $PM_{2.5}$-bound chlorinated paraffin exposure analysed by generalised linear mixed models, restricted cubic splines, WQS regression and quantile g-computation, plus mediation analysis. Per-IQR ΣCP exposure raised sleep-disorder odds with **ORs reported as a 1.08–3.20 range**; short-chain CPs were the dominant contributor; body weight mediated 3.40–35.67%; associations were stronger in girls and in children under 12. Two serious problems. The exposure is almost certainly modelled or area-assigned rather than personally measured — CP speciation on $PM_{2.5}$ is a specialist analysis not deployable at $n > 10^5$ — so the exposure contrast is spatial and confounded with everything else that varies across the PRD. And an OR range spanning 1.08–3.20 without a stated model set is uninterpretable: it could be lag structure, subgroup, or CP congener. Large n does not fix either.

doi: 10.3390/toxics14070607

Ion-Andrei et al. 2026 Biomedicines

Chronic air pollution exposure and the immunomolecular profile of lung cancer

Regional retrospective series in Constanţa County, Romania, linking ≈15-year historical monitor-based pollution exposure to tumour histology, PD-L1 expression and EGFR status, with ALK/KRAS described only. **Chronic NO_2 was associated with higher PD-L1 expression** (**OR 1.035,** $p = 0.017$), with PM_{10}, NO and NO_x trending in the same direction and an exploratory CO–EGFR signal. The idea — that ambient exposure shapes tumour immunophenotype and therefore immunotherapy eligibility — is genuinely interesting and underexplored. The execution does not support it: no CI or sample size in the abstract, no adjustment for smoking described, an exposure assigned from county monitors over 15 years with no residential-history weighting, and multiple correlated pollutants tested without correction. Hypothesis-generating at best, and the authors concede causality cannot be established.

doi: 10.3390/biomedicines14071615

Doğan Karataş et al. 2026 J Clin Med

Air pollution and sudden sensorineural hearing loss: nine-year retrospective study

223 SSHL patients (mean 52.7 ± 7.4 y) at a single ENT clinic in Sivas, Türkiye, 2016–2025, with monthly mean PM_{10}, $PM_{2.5}$, SO_2, NO_x, NO, CO and NO_2 from ministry monitoring stations and a "cumulative risk" computed by a public-health specialist. Post-treatment audiometric change was significant at all frequencies (greatest at 2 kHz) and no significant seasonal variation in case occurrence was found. The design cannot answer the question it poses: a case series with no control period and no non-case comparison cannot estimate an exposure–occurrence association, and pairing 223 events with monthly city-wide pollutant means guarantees exposure misclassification far exceeding any plausible effect. The audiometric outcomes are a treatment-response analysis with no pollution content. Included for completeness of the register.

doi: 10.3390/jcm15145687

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Lu	Subchronic $PM_{2.5}$ exposure exacerbates chronic stress-induced depression-like pathology in mice	*Environ Res*	Murine inhalation exposure	$PM_{2.5}$	China
Cardiovascular & metabolic					
Zhou	Air pollution and hypertension: umbrella review of systematic reviews and meta-analyses	*iScience*	Umbrella review / meta-analysis	$PM_{2.5}$ / PM_{10}	Global
Reproductive & developmental					
Liu	Deoxynivalenol, cytokines and particulate matter in human trophoblast cell models	*Int J Mol Sci*	In vitro (cell lines)	$PM_{2.5}$	Netherlands
Respiratory & allergic					
Chaiwong	$PM_{2.5}$ and systemic inflammatory biomarkers in elderly COPD patients versus healthy elderly	*Int J Mol Sci*	Panel study	$PM_{2.5}$	Thailand
Ghana AIRCLARE	Short-term ambient $PM_{2.5}$/PM_{10} and acute respiratory morbidity in Ghana: time-stratified case-crossover pilot	*Research Square (preprint)*	Case-crossover	$PM_{2.5}$ / PM_{10}	Ghana
Mechanistic toxicology					
Fakfum	Seasonal $PM_{2.5}$ exposure and plasma metabolome changes related to metabolic syndrome	*Toxics*	Repeated-measures biomarker	$PM_{2.5}$	Thailand
Lu	Toxicity of $PM_{2.5}$ from incomplete solid-fuel burning in Caenorhabditis elegans	*Toxics*	Controlled exposure (ecotox)	$PM_{2.5}$ (biomass)	China
Occupational & indoor					
Komonnithiphong	$PM_{2.5}$-bound PAHs and carcinogenic risk in charcoal barbecue (Moo Kratha) restaurants	*Toxics*	Field measurement	$PM_{2.5}$ components	Thailand
Subway PSD	Characterisation and source apportionment of PM and CO2 in subway stations with differing platform door forms	*J Environ Sci (China)*	Field measurement	$PM_{2.5}$ / PM_{10}	China
Sensing, forecasting & instrumentation					
Ainembabazi	Trends in atmospheric pollution research in Uganda, 1990-2025: a scoping review	*Toxics*	Scoping review	$PM_{2.5}$ / PM_{10}	Uganda
Wang	Joint monitoring and early warning of SARS-CoV-2 in outdoor $PM_{2.5}$ and wastewater	*Viruses*	Environmental surveillance campaign	$PM_{2.5}$ (bioaerosol)	China
Exposure assessment & modelling					
Chengdu NO3	Gas-phase reactions accelerate nitrate formation in spring and early summer in a Sichuan Basin megacity	*J Environ Sci (China)*	Supersite observation	$PM_{2.5}$ components	China
Pajak	Toxicological legacy of PAHs from an urban tire fire: soil contamination and cancer risk	*Toxics*	Dispersion model + soil sampling	PM_{10} / PAH	Poland
Burden, policy & mitigation					
Hu	Older-adult life expectancy in China: Bayesian mortality estimation and spatially heterogeneous determinants	*Front Public Health*	Bayesian + geospatial regression	PM_{10}	China
Sun	Clarifying causal and translational inference in studies of ambient $PM_{2.5}$ and lung cancer	*Br J Cancer*	Commentary / methods letter	$PM_{2.5}$	Global
Zhou	Sources of ultrafine particles during severe haze periods: a spatial Durbin analysis	*Toxics*	Spatial econometric model	Ultrafine particles	China
Other clinical endpoints					
Dogan Karatas	Air pollution and sudden sensorineural hearing loss: nine-year retrospective study	*J Clin Med*	Retrospective clinical series	PM_{10} / $PM_{2.5}$	Turkiye
He	$PM_{2.5}$-bound chlorinated paraffins and sleep disorders in youth, with body weight as mediator	*Toxics*	Cross-sectional (multistage)	$PM_{2.5}$ components	China
Ion-Andrei	Chronic air pollution exposure and the immunomolecular profile of lung cancer	*Biomedicines*	Retrospective cohort	PM_{10} (with NO2/NOx)	Romania

Provenance and method

Entry window 28 July 2026, continuous with the 24–27 July window of the previous issue; no gap. **Sources queried:** PubMed E-utilities against `[EDAT]` with two term sets (particle/health, 16 hits; sensing, calibration and exposure-assessment terms crossed with particle terms, 1 hit, itself a subset of the first); Europe PMC against `CREATION_DATE` 26–28 July (10 hits, 5 already published in the 27 July issue and removed by the dedup key); Crossref `from-index-date` (rejected as a source — Crossref re-indexes historical records continuously, so the filter returned works from 2007 onward and is unusable for entry-date windowing); OpenAlex `from_created_date` (**now returns HTTP 403, "Plan upgrade required" — this filter has moved behind a paid tier and the OpenAlex leg of the pipeline is broken until an alternative is found**); arXiv (the `submittedDate` range query returned an empty body on this run; the unfiltered relevance query surfaced nothing newer than 6 July). A supplementary web sweep for low-cost-sensor calibration work confirmed the entry-window gap is real rather than a query artefact.

Deduplication: PMID → lowercased DOI → SHA-1 of normalised title + first author + year, checked against `claude/state/seen.json` before any summarisation. **Screening:** 21 candidates, 19 in scope. Two exclusions logged to `claude/state/rejected.jsonl`: a deep-mine battery-fleet transferability review (diesel PM as motivating context only, no measurement, exposure model or health endpoint) and a water-transparency review (aquatic suspended particulate matter, not airborne). **Labels** for subtopic, design, particle metric and geography are single-label judgements from title, abstract and MeSH, not a controlled vocabulary. **Effect estimates** are transcribed verbatim; exposure contrasts are heterogeneous and estimates are not mutually comparable. One record is a **preprint** and is labelled as such throughout. All DOI links resolve to the publisher of record; abstracts remain the copyright of their respective publishers.

Correction, reissued 31 July 2026. As first published, the Sun et al. entry (PMID 42509466) carried `10.1016/j.lanwpc.2024.101106`, a DOI that PubMed's source metadata had registered incorrectly and which resolved to an unrelated *Lancet Reg Health West Pac* article. The record has since been corrected upstream and the link is now `10.1038/s41416-026-03555-2` (*Br J Cancer*). No other content in this issue has changed.